

Muawiya Amir

Software Engineer · Full Stack & Open Source

Multan, Pakistan | +92 329 7316882 | contactmuawia@gmail.com | github.com/Muawiya-contact | linkedin.com/in/contactmuawia

PROFESSIONAL SUMMARY

Software engineer with full-stack experience at SudoStudy and a strong open-source background. I built tools that help teachers create quizzes, grade work, and understand student progress. As a Triage Collaborator on [drt-hub/drt](#), I build connectors and observability tools and review community contributions. I study Artificial Intelligence and Mathematics and enjoy making complex systems useful to people.

TECHNICAL SKILLS

Languages: Python, C / C++, JavaScript, TypeScript, Java, SQL, x86_64 Assembly

AI / ML: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Computer Vision, NLP

Web & Backend: React, Node.js, Flask, REST APIs, HTML / CSS

Systems & DevOps: Git, GitHub Actions, Docker, CI/CD, pytest, JSON Schema, OpenTelemetry

CS Foundations: Operating Systems, Data Structures, Algorithms, Linear Algebra, Probability

PROFESSIONAL EXPERIENCE

Software Engineer (Full Stack), SudoStudy

Sep 2025 — Oct 2026

- Built teacher workflows for section-based quizzes, student assignment, Google Classroom publishing, and grade return.
- Developed scoring and performance reports with charts and PDF exports.
- Improved a Gemini study assistant with streaming responses and conversation history.
- Improved mobile teacher dashboards, fixed authentication and grading issues, and contributed to read-only teacher analytics that protects student contact details.

Open-Source Engineer & Triage Collaborator, [drt-hub/drt](#)

Apr 2026 — Present

- Promoted to Triage Collaborator after sustained, high-quality contributions; review pull requests, triage issues, and shape feature design with maintainers.
- Designed and built the dynamic connector registry that auto-discovers sources and destinations, removing hardcoded dispatch maps across the engine.
- Added the Jira connector (create/update), JSON Schema validation for config files, and data-quality tests (freshness, unique, accepted_values).
- Shipped OpenTelemetry tracing and metrics with lazy, idempotent providers (+680 lines) and dry-run row-count diffs for safe sync previews.

PROJECTS

Linear Algebra Library — A Python numerical library built only from primitive types — no math libraries — implementing core linear-algebra routines by hand.

SudoStudy — Teacher tools for quizzes, Google Classroom publishing, grading, and student performance reports on an education platform.

BSAI University Projects (NFC-Projects) — Structured collection of semester-wise coursework and systems projects for my BS in Artificial Intelligence at NFC IET Multan. Includes data structures, algorithms, IoT simulators, search engine implementations, and AI academic tools.

Diskern — Intelligent, read-only disk cleaner and analyzer that scans local file systems to determine file safety before freeing space. Built with a deterministic rules engine in Rust ([crates/diskern-core](#)) and a Tauri v2 + React desktop interface.

OPEN-SOURCE CONTRIBUTIONS

- [drt-hub/drt](#)** — Triage Collaborator on a Python Reverse ETL engine. Designed the dynamic connector registry, added connectors and config validation, and shipped observability support.
- [apache/hugegraph-ai](#)** — Upgraded the Python client CI from 1.3.0 to 1.7.0 and migrated from manual Docker runs to GitHub service containers with health checks.
- [linuxfoundation/crowd.dev](#)** — Audited and corrected outdated technical documentation, Node version, and repository URLs across the project.

EDUCATION

BS Artificial Intelligence

NFC Institute of Engineering & Technology, Multan

BS Mathematics

Virtual University of Pakistan

CERTIFICATIONS & COMPETITIVE PROGRAMMING

- Applied Machine Learning — Verified
- Python AI Developer — Mimo
- Google Cloud Specialization — NAVTTC (ongoing)
- 400+ problems solved on LeetCode, focused on DSA & algorithm optimization.